Ondřej Kubů

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PROFILE

Research mathematician (Severo Ochoa postdoc, ICMAT-CSIC Madrid) moving into technical AI safety or governance. Doctoral training in mathematical physics: differential geometry and integrable systems, with substantial graduate work in operator and spectral theory. 8 peer-reviewed articles, one further article in press, and 1 preprint under review.

AI SAFETY

Public writing: opened a series profiling AI-safety orgs for mathematicians moving into the field with *ARC's research agenda: solid mathematics, an unclear safety case* (Jul 2026).

Volunteer, PauseAI CZ, z. s. (2026–present). Author of the Czech-language weekly newsletter *Pozastavení nad AI*, synthesizing frontier-AI developments and their EU and Czech policy implications for a non-specialist audience.

AGI Strategy course, BlueDot Impact (Jul 2026), certificate. Final project on compute-governance verification through hardware-enabled mechanisms.

Independent study of formal impossibility results in AI control (Alfonseca et al. on containment, Yampolskiy on controllability): checking which proofs hold and which applications of Rice's theorem go through.

Evaluation of theory-first safety agendas, ongoing: worked through ARC's mechanistic-safety programme (incl. Winer's account, Matolcsi's obstacles, and Wu et al. 2026 on cumulant-propagation estimators) and Bengio's Scientist AI when choosing a research direction.

Giving What We Can 10% Pledge. Committed to donating 10% of lifetime income to effective charities.

EMPLOYMENT

Dec 2024 – Nov 2026	**Postdoc (Severo Ochoa)**, Instituto de Ciencias Matemáticas (ICMAT), CSIC, Madrid. Haantjes algebras for separation of variables of (super)integrable systems. **Supervisor:** P. Tempesta
Sep 2024 – Nov 2024	**Research Associate**, FNSPE, Czech Technical University in Prague
2021 – 2024	**Assistant lecturer**, Dept. of Physics, FNSPE CTU. Seminars in Theoretical Physics 2, Lie Algebras and Groups, and Geometrical Methods in Physics.

EDUCATION

2020 – 2024	**Ph. D.**, FNSPE CTU Prague, Mathematical Engineering / Mathematical Physics. **Thesis:** *Integrability and superintegrability in the presence of magnetic fields: separable and nonseparable systems.* **Supervisor:** L. Šnobl; **Co-supervisor:** A. Marchesiello. *Rector's Award for Excellent Doctoral Thesis (2025)*. **Grade:** 1.00 GPA (Excellent); oral defence Grade 1.
2018 – 2020	**Master's Degree with Honors (Ing.)**, FNSPE CTU, Mathematical Physics. **Supervisor:** L. Šnobl; **Consultant:** P. Winternitz. **Grade:** A (GPA 1.31).
2015 – 2018	**Bachelor's Degree with Honors**, FNSPE CTU. **Grade:** A (GPA 1.41).

RESEARCH VISITS

ERASMUS+ research exchanges during studies: Universidad Complutense de Madrid, Spring 2023 (P. Tempesta); Université de Montréal, Fall 2019 (P. Winternitz).

AWARDS

Rector's Award for Excellent Doctoral Thesis, CTU (2025); **Josef Hlávka Award** for the best students and graduates of public universities in Prague (2023); **Milan Odehnal Award** (2024), honorable mention, Czech Physical Society; **Dean's Award** for the 2nd best Master Thesis (2021).

SKILLS

Software: Maple, LaTeX; author of an open Maple package for Nijenhuis/Haantjes torsion computation, github.com/OndrejKubu/NijenhuisHaantjesTorsion (2026). Basics of C++; daily use of AI coding assistants in research.

Coursework: Doctoral courses, all graded 1 (excellent): Schrödinger operators; geometrical aspects of spectral theory; operator semigroups; classification and identification of Lie algebras; solvable models of mathematical physics.

Service: 4 peer reviews for *Journal of Physics A*.

Languages: Czech (native), English (C1–C2), German (B2/C1), French (B2), Spanish (A2).

PUBLICATIONS

Full list: ORCID 0000-0001-7463-5680.

Preprint

O. Kubů, D. Latini. **Haantjes algebras, Zernike system and separation of variables**, under review. arXiv:2605.23748 (2026)

Journal articles

O. Kubů, P. Tempesta. **Stäckel and Eisenhart geometries, symplectic-Haantjes manifolds and gravitation**, accepted, *Proc. R. Soc. A*; in press. DOI: 10.1098/rspa.2025.1103. arXiv:2509.19950

O. Kubů, D. Reyes, P. Tempesta, G. Tondo. Hamiltonian integrable systems in a magnetic field and symplectic-Haantjes geometry. *Proc. R. Soc. A* **480** (2024) 20240076

O. Kubů, A. Marchesiello, L. Šnobl. Integrable systems with velocity-dependent potentials: generalized parabolic cylindrical case. *J. Phys. A* **57** (2024) 235203

O. Kubů, A. Marchesiello, L. Šnobl. New classes of quadratically integrable systems in magnetic fields: the generalized cylindrical and spherical cases. *Ann. Phys.* **451** (2023) 169264

O. Kubů, L. Šnobl. Cylindrical first-order superintegrability with complex magnetic fields. *J. Math. Phys.* **64** (2023) 062101. *(Non-self-adjoint Hamiltonians: pseudo-Hermiticity without quasi-Hermiticity.)*

M. F. Hoque, O. Kubů, A. Marchesiello, L. Šnobl. New classes of quadratically integrable systems with velocity dependent potentials: non-subgroup type cases. *Eur. Phys. J. Plus* **138** (2023) 845

O. Kubů, A. Marchesiello, L. Šnobl. Superintegrability of separable systems with magnetic field: the cylindrical case. *J. Phys. A* **54** (2021) 425204

S. Bertrand, O. Kubů, L. Šnobl. On superintegrability of 3D axially-symmetric non-subgroup-type systems with magnetic fields. *J. Phys. A* **54** (2021) 015201

O. Kubů, L. Šnobl. Superintegrability and time-dependent integrals. *Archivum Math.* **55** (2019) 309–318

SELECTED TALKS

Invited, *Geometry and Integrable Systems*, La Trobe University, Melbourne (Feb 2026); **Invited**, *Superintegrability, Exact-Solvability, and Representation Theory* (Nov 2024); Seminar, Università degli Studi di Milano (Feb 2026); *XXXIII International Fall Workshop on Geometry and Physics* (Sep 2025). Full list on the website.